

KILLARNEY, ON, Canada

WMO: 714600

Lat: 45.9682N

Lon: 81.4888W

Elev: 644

StdP: 14.36

Time Zone: -5.00 (NAE)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-10.8	-5.4	-23.7	1.5	-7.4	-18.1	2.1	-2.0	34.7	23.6	30.1	25.0	7.1	40	0.624

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.2	76.9	64.5	74.4	63.7	72.3	63.5	69.0	72.8	67.6	71.1	66.3	70.0	11.8	280

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
67.6	103.9	71.0	66.2	99.0	70.0	64.8	94.0	68.8	33.5	72.7	32.4	71.2	31.4	70.3	76.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%		2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
Min		Max		Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)		(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
28.3	24.6	21.2	DB	-17.1	83.2	5.6	3.6	-21.2	85.8	-24.5	87.9	-27.6	90.0	-31.7	92.6
			WB	-17.7	71.4	5.5	2.8	-21.7	73.4	-24.9	75.0	-27.9	76.6	-31.9	78.7

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
Temperatures, Degree-Days and Degree-Hours	DBAvg	42.6	16.2	17.7	26.2	38.4	50.6	60.8	66.1	65.7	59.3	47.6	36.0	24.5	
	DBStd	19.45	12.51	10.56	9.60	7.12	6.23	5.57	4.47	4.32	6.19	7.21	8.48	10.45	
	HDD50	4464	1049	903	737	353	67	2	0	0	7	134	420	792	
	HDD65	8367	1514	1323	1202	799	448	146	41	43	185	540	869	1257	
	CDD50	1749	0	0	0	4	86	327	498	487	287	59	1	0	
	CDD65	176	0	0	0	0	1	21	74	65	15	0	0	0	
	CDH74	271	0	0	0	0	3	46	133	83	6	0	0	0	
	CDH80	26	0	0	0	0	0	5	12	9	0	0	0	0	
Wind	WSAvg	9.4	10.0	10.0	9.5	9.5	8.4	7.7	8.0	8.1	8.9	10.9	11.5	10.9	
Precipitation	PrecAvg	39.0	2.9	2.1	2.2	3.2	3.5	3.3	3.3	3.7	3.9	4.3	3.5	3.0	
	PrecMax	46.1	4.7	4.4	5.5	5.8	6.1	5.6	5.4	6.6	7.6	7.0	5.4	6.1	
	PrecMin	31.9	1.5	0.8	0.3	1.2	1.3	1.1	1.5	1.9	1.5	1.9	1.6	1.3	
	PrecStd	3.8	0.8	0.9	1.1	1.1	1.2	1.1	1.1	1.1	1.4	1.4	1.0	0.9	
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	41.3	41.5	49.4	59.9	71.7	79.1	81.4	80.4	74.6	65.6	53.8	46.7	
		MCWB	39.7	38.4	40.0	45.9	57.4	62.0	67.5	68.2	66.0	59.9	49.2	43.8	
	2%	DB	38.3	37.4	44.1	54.5	65.8	74.3	77.3	76.1	71.5	62.6	51.1	43.0	
		MCWB	36.4	34.8	37.1	43.4	55.5	61.3	65.1	65.4	65.2	57.9	47.8	40.5	
	5%	DB	35.7	34.8	40.4	50.7	62.0	70.8	74.8	73.5	69.5	60.5	49.1	40.3	
		MCWB	33.8	32.8	35.1	41.7	53.1	60.4	63.5	64.4	64.0	56.2	46.3	38.1	
	10%	DB	33.4	32.7	37.6	47.3	59.3	68.2	72.6	71.7	67.7	58.2	47.3	37.7	
		MCWB	31.7	30.9	33.4	40.1	51.5	60.0	63.3	64.2	62.9	54.2	44.5	35.5	
	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	40.5	39.1	43.8	49.4	62.0	67.7	71.1	71.2	69.3	61.9	50.9	44.8
			MCDB	40.9	40.5	45.6	55.3	67.5	72.4	77.1	76.4	71.7	64.3	52.7	46.2
2%		WB	36.6	35.3	38.7	45.9	58.1	65.4	69.0	69.1	67.3	59.5	48.6	41.0	
		MCDB	37.7	36.5	42.1	50.8	62.6	69.5	73.5	72.6	69.8	61.8	50.3	42.4	
5%		WB	34.2	33.2	36.0	43.4	55.5	63.9	67.4	67.8	65.6	57.1	46.9	38.3	
		MCDB	35.6	34.5	39.0	48.5	59.7	67.8	71.6	70.9	68.2	59.6	48.9	40.0	
10%		WB	31.8	31.0	34.1	41.2	53.0	62.2	65.9	66.5	64.0	54.8	44.7	35.8	
		MCDB	33.4	32.6	37.1	46.2	57.7	66.5	70.3	70.1	67.0	57.8	46.9	37.5	
Mean Daily Temperature Range	5% DB	MDBR	13.6	14.1	12.9	11.4	11.8	10.8	10.2	9.6	9.9	9.4	9.6	11.1	
		MCDBR	10.9	12.3	12.4	15.7	15.4	14.1	12.8	11.5	10.5	9.9	10.1	10.5	
		MCWBR	10.6	11.7	8.9	9.4	8.3	7.3	6.9	7.2	7.8	8.7	10.2	10.3	
	5% WB	MCDBR	10.9	12.0	10.9	13.1	13.0	10.9	10.6	8.9	9.0	9.0	9.8	10.3	
		MCWBR	10.9	11.8	9.3	9.7	9.0	7.8	7.3	7.1	8.3	9.0	10.4	10.5	
Clear-Sky Solar Irradiance	taub	0.284	0.288	0.309	0.349	0.366	0.396	0.402	0.392	0.364	0.343	0.311	0.276		
	taud	2.371	2.323	2.380	2.361	2.405	2.340	2.355	2.376	2.439	2.469	2.505	2.471		
	Ebn at Noon	260	283	292	288	286	276	273	272	270	260	250	254		
	Edh at Noon	23	30	32	36	36	39	38	36	31	26	21	19		
All-Sky Solar Radiation	RadAvg	384	682	1140	1462	1735	1876	1899	1604	1232	729	396	282		
	RadStd	51	71	106	127	162	142	104	94	121	86	55	26		

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD50	HDD65	CDD50	CDD65
(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (18 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page